Since OP is perpendicular to the line (6.25)

$$m \times \left(\frac{k-0}{h-0}\right) = -1 \Rightarrow m = -\frac{h}{k}$$

Also $P(h, k)$ lies on (6.25) $\Rightarrow k - 8 = -\frac{h}{k}(h - 6)$

$$\Rightarrow k\,(k-8) = -h(h-6) \Rightarrow h^2 + k^2 - 6h - 8k = 0$$

Locus of $P(h, k)$ is $x^2 + y^2 - 6x - 8y = 0$

This result shows from the fact that the angle in a semi circle is a right angle.

6.4.5 Family of lines

All lines follow a specific condition are called a family of lines. The following example shows some families of straight lines.(where $m.h,$ and k are arbitrary constants).

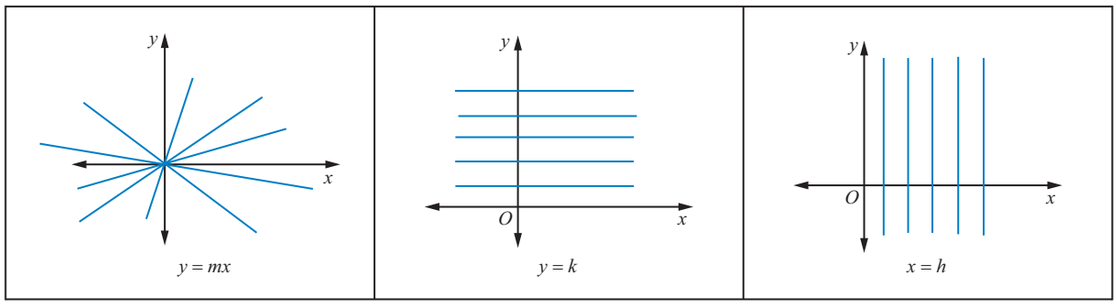


Figure 6.39

It may seem that the equation of a straight line $ax + by + c = 0$ contains three arbitrary constants. In fact, it is not so. On dividing it by b (or a, whichever is non-zero), we get

$$\frac{a}{b}x + y + \frac{c}{b} = 0, \quad \text{which can be written as}$$
$$Ax + y + C = 0 \quad \text{where} \quad A = \frac{a}{b} \text{ and } C = \frac{c}{b}.$$

The above equation can be written as slope and intercept form.

It follows that the equation of a straight line contains two arbitrary constants, and the number of these arbitrary constants cannot be decreased further. Thus, the equation of every straight line contains two arbitrary constants; consequently, two conditions are needed to determine the equation of a straight line uniquely.

One condition yields a linear relation among two arbitrary constants and hence each arbitrary constant determines the other. Therefore, the lines which satisfy one condition contain a **single arbitrary constant.** Such a system of lines is called **one parameter family of lines** and the unknown arbitrary constant is called, the **parameter.**

Let us now discuss the three types of families of straight lines, using $y = mx + b$. First two types are one parameter families and third one is two parameters families

(i) when m is arbitrary and b is a fixed constant.
(ii) when b is arbitrary and m is a fixed constant.
(iii) when both m and b are arbitrary